

Is Time Dilation ‘Real’? Einstein and the Transverse Doppler Effect

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Abstract

From early on, the uncertainty relations were understood to preclude exact predictions of trajectories, while apparently still allowing their retrospective reconstruction with arbitrarily precise position and momentum values. In his Chicago lectures, Heisenberg treated the physical reality of such reconstructed trajectories as a ‘matter of taste’. In 1934, Popper seized on this concession and proposed a thought experiment that, by using conservation laws, sought to turn a sharp *trajectory reconstruction* into an exact trajectory prediction. Heisenberg and his assistants sharply criticized the proposal. This paper argues that in early 1935 Hermann, drawing on a structurally similar thought experiment, identified Popper’s deeper confusion between context-independent trajectory reconstruction and context-dependent *measurement reconstruction*. It shows that the twenty-two-year-old von Weizsäcker, then Heisenberg’s assistant, played a unique role both in criticizing Popper’s arrangement and in providing the thought experiment on which Hermann relied. The paper concludes that, on the eve of the EPR debate, the Hermann–Popper controversy came close to raising the question of whether one may legitimately combine context-dependent attributions of position and momentum into a description of the underlying reality.

Keywords: Karl Popper, Grete Hermann, history of quantum mechanics, complementarity, thought experiments

Acknowledgements

I am grateful to Andrea Reichenberger for generously sharing unpublished archival material by Grete Hermann.

Ethical Statement

- Funding: None
- Conflict of Interest: None declared
- Ethical approval: Not required
- Informed consent: Not required
- Author contribution: I am the sole author of this paper